Agentic Intelligence for Automated Document Extraction and Analysis

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Project Overview :

The project presents a multi-agent architecture built using Agentic AI principles, aimed at intelligent processing of PDF documents. It includes a centralized manager agent and three specialized sub-agents working collaboratively.

The parameters include :

- The system is composed of a **Manager Agent** and three specialized **Sub-agents** :
- **Librarian:** Extracts structured content from PDFs.
- **Artist:** Analyzes visual styling (fonts, colors.
- **Bookkeeper:** Loads CSV files into SQLite database and creates requested graphs.

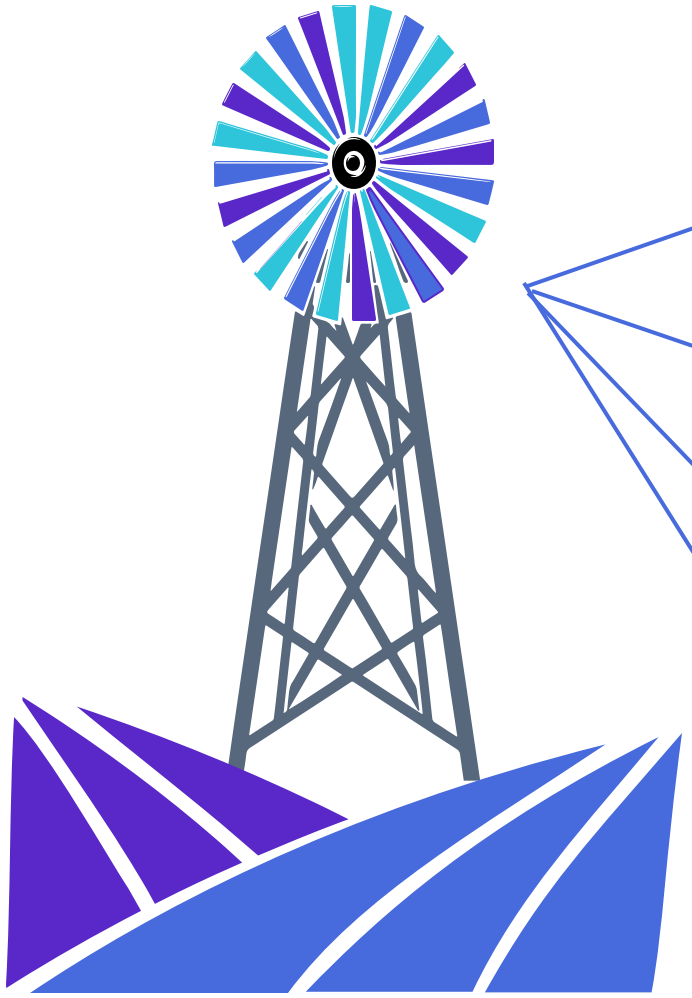
Project Overview :

Key capabilities :

- The system uses **agent-based reasoning** to divide and manage complex document **automating PDF analysis**, transforming unstructured documents into actionable data.

- Manager Agent acts as the **central coordinator**, handling user queries and delegating tasks to sub-agents.flow.

Technology Stack :

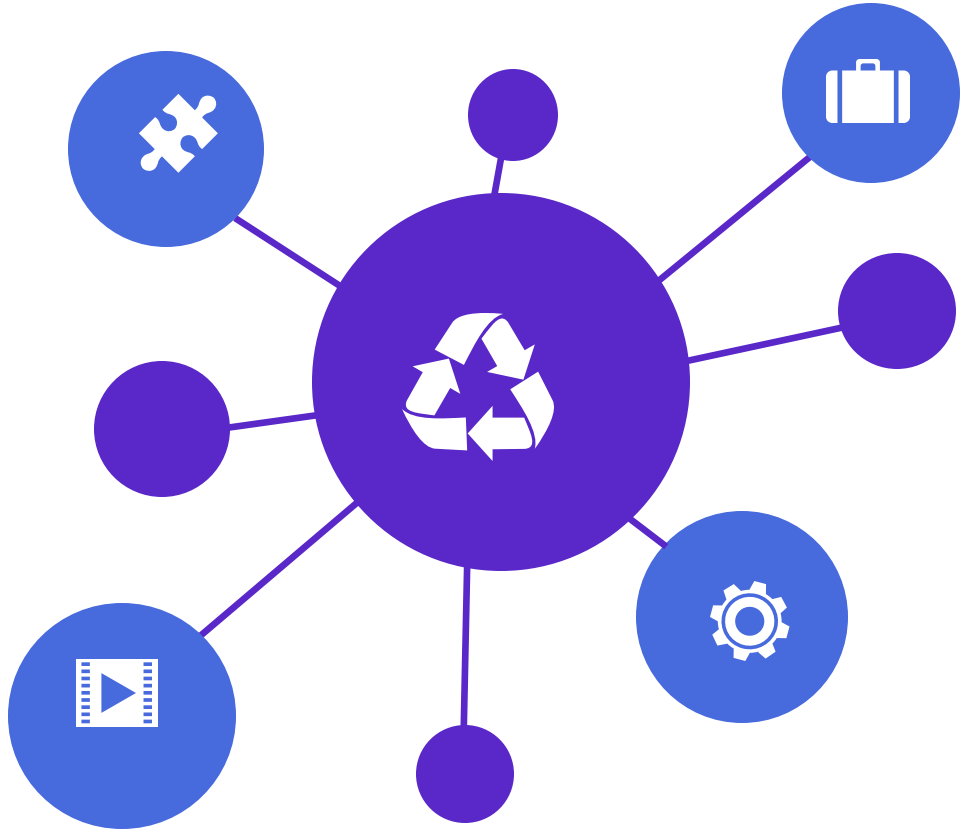


- **Fast API** — Lightweight, asynchronous web framework to build backend.
- **Postman** — Acts as client interface to test and send HTTP requests.
- **Cloudinary** — Cloud-based image management platform.
- **Streamlit** — Lightweight Python-based web UI.

Manager Agent :

Handles User Input :

- Receives uploaded PDFs and chat-based queries via FastAPI endpoints.
- Acts as the central interface between the user and the agent system.



Decides Query Routing :

- Analyzes the user's query to determine which sub-agent (Librarian, Artist, or Bookkeeper) is most appropriate for handling it.
- Uses rule-based logic or keyword mapping to decide routing.

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Manager Agent :

Inter-Agent Communication :

- Sends the query and any necessary data to the selected sub-agent.
- Waits for the response and logs the result.

Validates Agent Responses :

- The Manager Agent validates if the sub-agent's response matches the query intent.
- If relevant, it returns it to the user, otherwise, it reroutes the query to another sub-agent.

Manager Agent – Workflow :

Input Stage:

Accepts:

- POST /upload_pdf: Stores the PDF.
- POST /chat: Processes a query about the uploaded PDF.

Decision Stage:

- Identifies the nature of the task: extraction, visual analysis, or data formatting.

Delegation Stage:

- Sends the request to the appropriate sub-agent.
- Supplies relevant file paths, table structures.

Evaluation Stage:

- Validates sub-agent's response.
- If necessary, retries with a different sub-agent.

Output Stage:

- Sends response to user via API.
- Asks user: "Do you need further assistance?"

User Interaction – Workflow :

Step 1: User uploads a PDF using the POST /upload_pdf endpoint in Postman.

Step 2: System stores the PDF temporarily for processing.

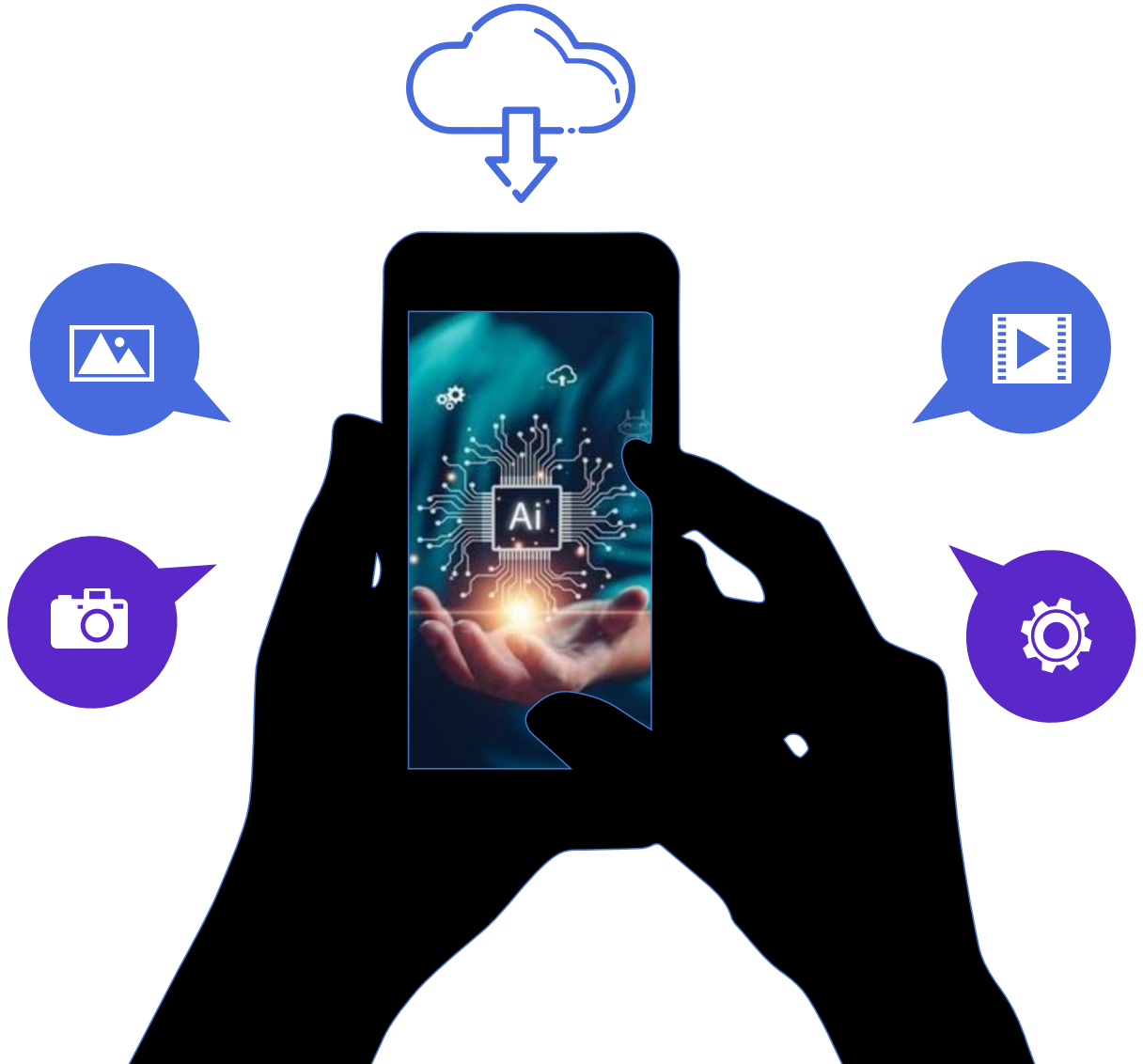
Step 3: User sends a query through the POST /chat endpoint.

Step 4: Manager Agent receives the query, initiates processing, and returns the response.

Librarian Agent :

Introduction to Sub-Agent 1 :

- The **Librarian Agent** is responsible for extracting structured information from uploaded PDF documents.
- It acts as the system's **data acquisition component**, parsing the content into three main types—text, images, and tables.
- The extracted data is formatted into JSON, enabling smooth communication.
- This agent ensures that raw document content is efficiently converted into usable digital data.



Librarian Agent – Architecture :

Input Interface :

Receives a **file path** uploaded PDF from the Manager Agent.

PDF Parsing Layer :

Utilizes a PDF parser and extracts :
-> Text
-> Images
-> Tables

Image Handling Module :

Extracted images are saved to Cloudinary API and URL is stored in JSON file.

Table Structuring:

Uses layout cues to reconstruct tables and output is in structured JSON format.

Output Interface :

Returns three separate JSONS. Outputs are sent back to Manager Agent for further analysis.

Artist Agent :

Introduction to Sub-Agent 2 :

Responsible for **visual content analysis** of images extracted from user-submitted PDFs..

Accessible via a FastAPI POST endpoint (/analyze-image) that accepts image files and optional text queries.

Performs tasks such as:

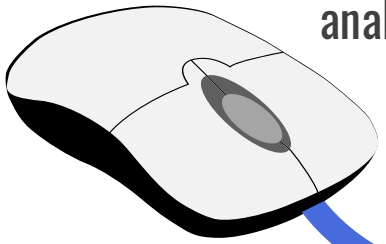
- Color palette analysis
- Font style and family detection
- Logo detection and metadata extraction

Uses **GPT-4o** for interpreting visual features and generating structured outputs.

Artist Agent - Architecture :

Input Stage :

Accepts individual image files and Optionalizes user query for focused analysis



Processing Layer :

FastAPI interface receives and processes requests and Analyzes:

- Dominant color hex codes
- Font families and styles
- Logos and associated metadata

Output & Logging :

- Logs results including color, font, and logo information.
- Structured data is returned via FastAPI.

Data Management :

- Stores image data and analysis results in a vector database.
- Uses FAISS indexing.

Final Output :

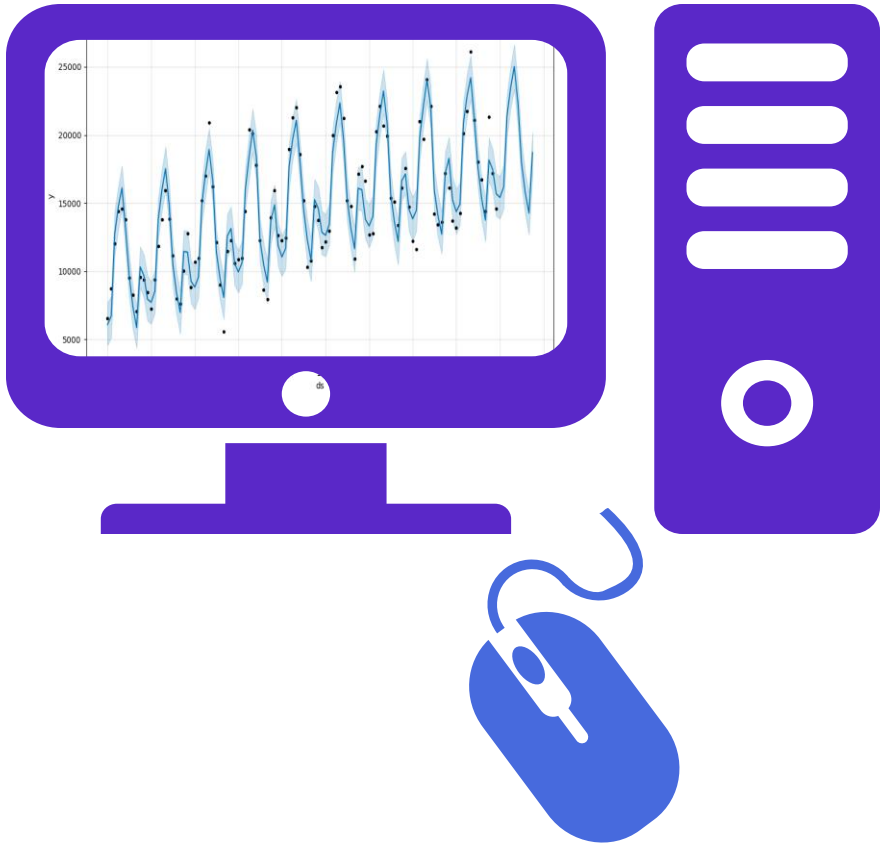
Sends analysis back to the Manager Agent.

Book-Keeper Agent :

Introduction to Sub-Agent 3 :

Purpose:

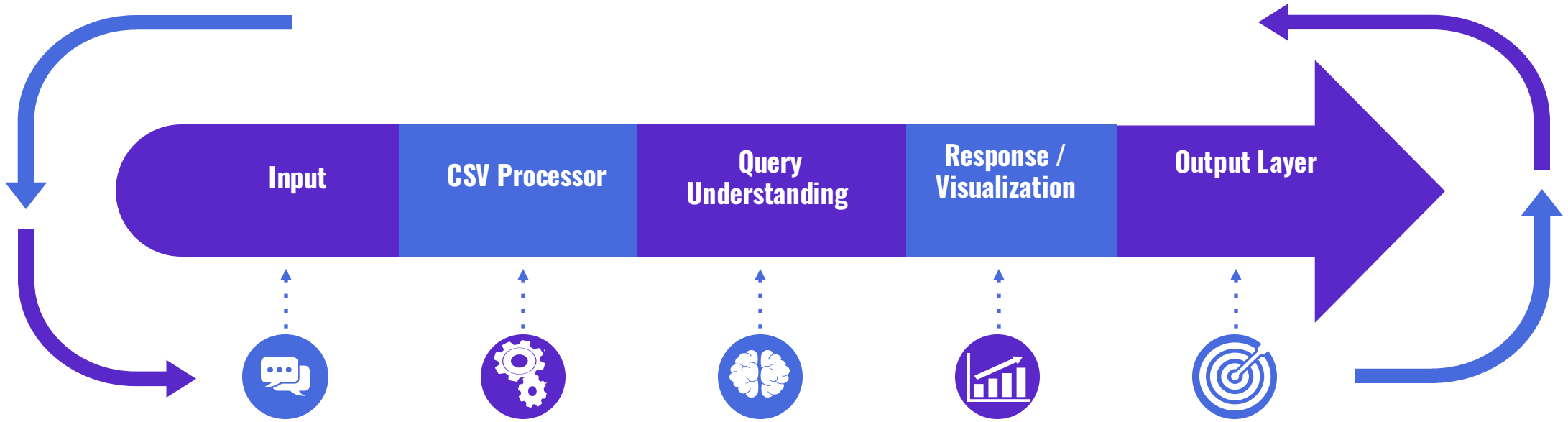
- Acts as an Intelligent CSV Analyzer Agent.
- Handles uploaded CSV files for data inspection, Q&A, and visualization.



Key Responsibilities:

- Uploads data to local database.
- Answers natural language queries such as:
- “What is the average temperature?”
- “List average snowfall by year.”
- Generates smart visualizations:
- Plots time-series graph.
- Line charts
- Bar plots

Book-Keeper - Architecture :



CSV file from user

Query string in natural language.

Saves data to local SQLite database. Summarizes dataset

Uses OpenAI agent to interpret the question.

Maps query to data operations (aggregation, filtering, plotting).

Generates graphs using Matplotlib

Supports advanced plots like line, bar, and scatter graphs.

Textual answers (summaries).

Embedded visualizations.

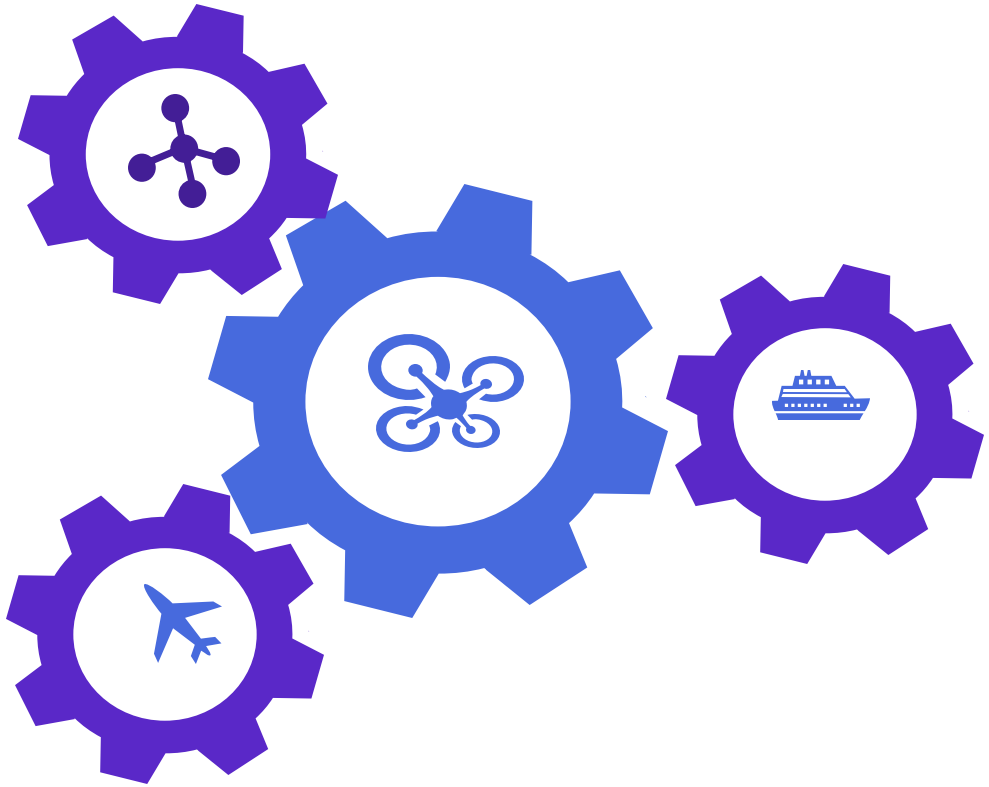
Output is routed back to the Flask api for client display.

Architecture & Interaction :

API Architecture :

FastAPI Endpoints:

- POST /upload_pdf – Upload a PDF.
- POST /chat – Submit queries.



API Interaction :

- Upload PDF → Manager distributes to agents.
- Chat with system → Manager routes to agents and compiles responses.

System Logic :

System Response Logic :

Manager Agent’s Role in Validation:

- Evaluates if the sub-agent's answer matches the query intent.
- **If relevant:** response is sent to the user.
- **If irrelevant:** query is redirected to another agent.

Goal:

- Ensure accurate and contextually relevant output every time.

Result:

- Only the best-fit answer is returned to the user after internal verification.

Impact :

Project Impact :

- Automates tedious document analysis tasks.

Can be applied in:

- Legal: Summarize case documents.
- Medical: Parse patient reports or scans.
- Academia: Process research PDFs and extract visuals/tables.

- Reduces human error and increases throughput in data handling.

Future Enhancements :

Future Enhancements :

- Expand file format support to DOCX, XLSX, scanned images (OCR).
- Incorporate Natural Language Understanding (NLU) to interpret queries more contextually.
- Develop a multi-user collaborative interface (e.g., web dashboard).
- Introduce learning-based routing (use ML to predict which agent should handle the query).

Conclusion :

-->Successfully implemented a multi-agent system using Agentic AI principles.

-->The system handles PDF processing, content extraction, visual analysis, and data visualization through coordinated sub-agents.

-->Manager Agent plays a central role in routing user queries and validating responses from sub-agents.

-->Achieved smooth integration with tools like FastAPI, Postman, and Cloudinary.

Q & A :

Q1: Why did you choose FastAPI over other web frameworks?

Q2 : How does the Manager Agent decide which sub-agent should handle a user query?

Q3: How is Cloudinary integrated, and what is its role in the system?

Q4: How does the system ensure data privacy and security?

Q5: What future improvements are planned for intelligent query routing?

Q6: Can the system be deployed in a private cloud or local environment?

THANK YOU

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